Week 9: Troubleshooting Reports and Dialogues – Writing and Discussing About Problems and Solutions

Unit ILO:
Describe technical problems clearly and propose appropriate solutions in writing using structured language and engineering terminology.

Unit ILO: Engage in simulated troubleshooting conversations using clear problem-solving language.

Session 1: Analyzing Troubleshooting Reports and Common Errors

Session ILO:
Identify the structure and typical language used in troubleshooting reports through analysis of authentic or simulated examples.

Learning Task 1: Troubleshooting Log Analysis

Learning Task 2: Problem–Cause–Solution Matching

Performance Task (Part 1): Troubleshooting Brainstorm

Language Focus:

- Causatives (“was caused by,” “resulted in”)
- Conditionals (“If the cable is unplugged, the system will not boot.”)
- Action verbs (“adjust,” “replace,” “reset,” “inspect”)

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Learning Task 1: Troubleshooting Log Analysis
Look at 2 sample troubleshooting logs (simplified versions from engineering settings).

Identify:

- Where the problem is stated
- How the cause is described
- What solution was attempted

    - What the result was

Look for layout elements:

- Date
- Technician
- Problem
- Action Taken
- Result/Follow-Up

Learning Task 2: Problem–Cause–Solution Matching

Match a set of 6–8 mini case descriptions with corresponding causes and solutions

(e.g., “Machine failed to initialize” ⇨ “Sensor cable was loose” ⇨ “Reseated the cable and restarted system”).

→ Emphasize logical connections using signal words (“due to,” “resulted in,” “was caused by…”).

Performance Task (Part 1): Troubleshooting Brainstorm

Create a simple troubleshooting scenario from **past lab work** or **hypothetical situations**:

- Equipment or tool involved
- Type of error or symptom
- Possible root cause
- Solution(s) attempted or recommended

Session 2: Writing a Troubleshooting Report

Session ILO:
Write a complete, structured troubleshooting report using problem–solution logic and accurate engineering language.

Learning Task: Mini-Lesson – Troubleshooting Report Tone and Format

Performance Task:
Students write a 180–250 word troubleshooting report based on their Session 1 brainstorm.

Language Focus:

- Problem language ("The system failed to respond…")
- sequence of actions
- cause-and-effect language ("This issue was likely due to…")
- polite recommendation phrasing ("It is advised to…")

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Learning Task: Mini-Lesson – Troubleshooting Report Tone and Format

View report structure on projector or board:

- o Header (Date, Location, Technician Name)
- o Problem description
- o Observed symptoms
- o Cause analysis
- o Corrective actions taken
- o Result/Outcome
- o Optional: Recommendation for future

Performance Task:
Write a 180–250 word troubleshooting report based on their Session 1 brainstorm.

→ Peer partners use a checklist to review:

- Clear problem identification
- Logical solution steps
- Use of conditionals or causatives
- Final result stated clearly